Understanding the Transition from Memorization to Generalization in Diffusion Models for Cosmology

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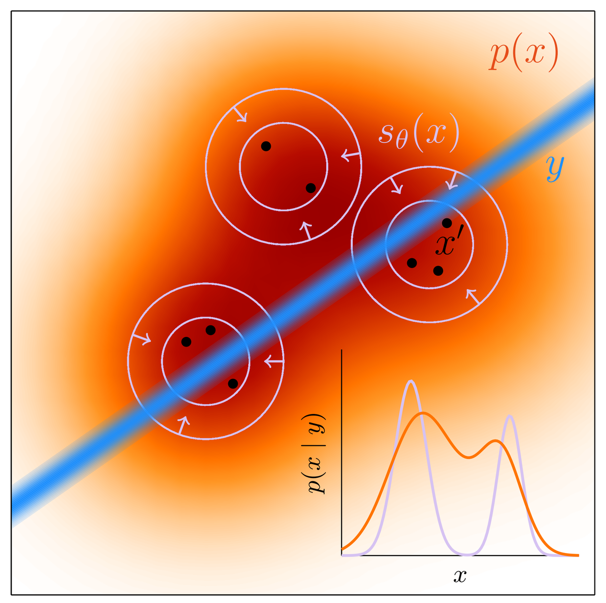
The question

Diffusion models are attractive as fast priors for cosmological fields, but inference needs more than visual realism: samples must be new, conditioned, and statistically faithful. On CAMELS HI maps we ask:

1. **Memorization:** when do generated maps stop matching individual training slices?
2. **Conditioning:** does changing $\theta_{\rm ask}$ change the recovered cosmology?
3. **Fidelity:** do novel maps retain the one-point PDF and power spectrum?

Toy picture: if the learned prior covers only training islands, the posterior can be sharp but displaced from the true distribution.



Data & models

- **CAMELS** (IllustrisTNG) LH set: 1000 simulations → 128 slices each → 128,000 candidate 128×128 HI maps.
- Six-parameter conditioning: $\boldsymbol{\theta} = (\Omega_m, \sigma_8, A_{\rm SN1}, A_{\rm AGN1}, A_{\rm SN2}, A_{\rm AGN2})$.
- **Diffusion objective.** We corrupt a real map with noise and train a UNet to predict the velocity target at every noise level, with the cosmology $\boldsymbol{\theta}$ as conditioning.

$$x_t = \sqrt{\bar\alpha_t}\,x_0 + \sqrt{1-\bar\alpha_t}\,\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),$$
$$\mathcal{L}_{\rm diff} = \mathbb{E}_{x_0,\epsilon,t}\,\|f_\theta(x_t, t, \boldsymbol{\theta}) - \mathrm{target}\|_2^2 .$$

Here the config uses `v_prediction`: $\mathrm{target} = v_t = \sqrt{\bar\alpha_t}\epsilon - \sqrt{1-\bar\alpha_t}x_0$. Ho et al. 2020 (DDPM).

- **Sampling**: 50-step DPM-Solver, $8\times$ faster than the 500-step DDPM baseline at matched quality.
- **Training-set sweep**: $N_{2D} = 2^6 \to 2^{15}$ fields, matched optimizer-update budget; runs on U-M Great Lakes HPC.

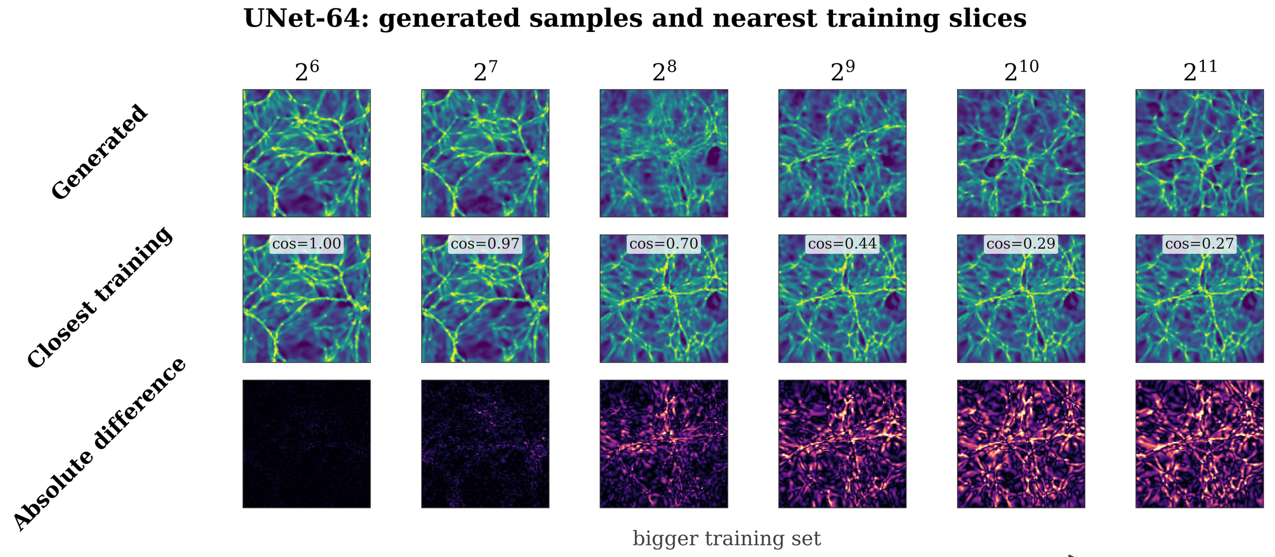
Detecting memorization

For each generated map x_j, compute its similarity to every training map y_i and keep the closest match:

$$s_j = \max_i \, \mathrm{sim}(x_j, y_i)$$

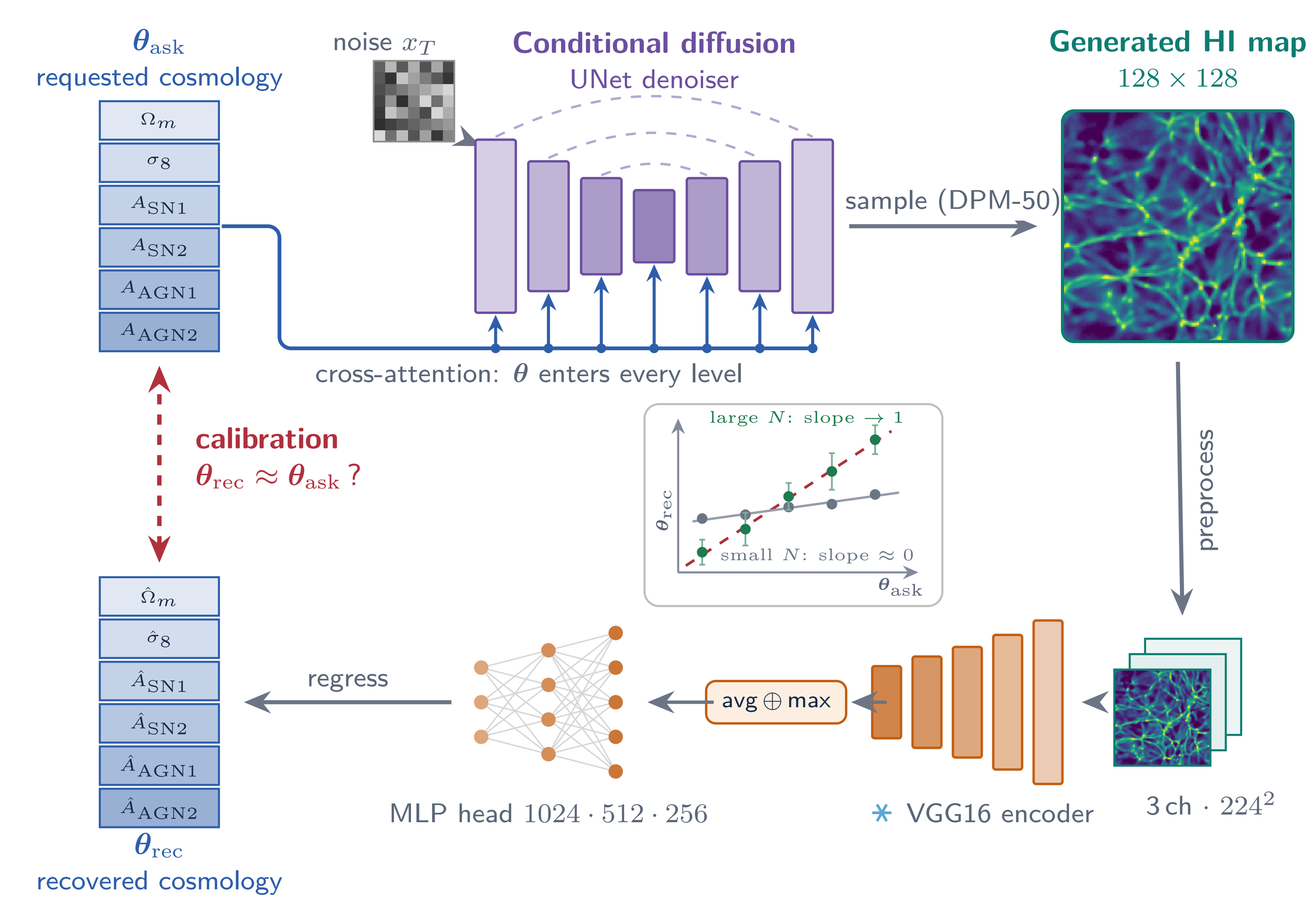
A high s_j means x_j is nearly a copy of one training map. We measure sim in pixel, PCA, and SSCD space. Two independently trained models should agree statistically while no individual map repeats; repeated maps mean both copied the training set.

UNet-64: generated samples and nearest training slices

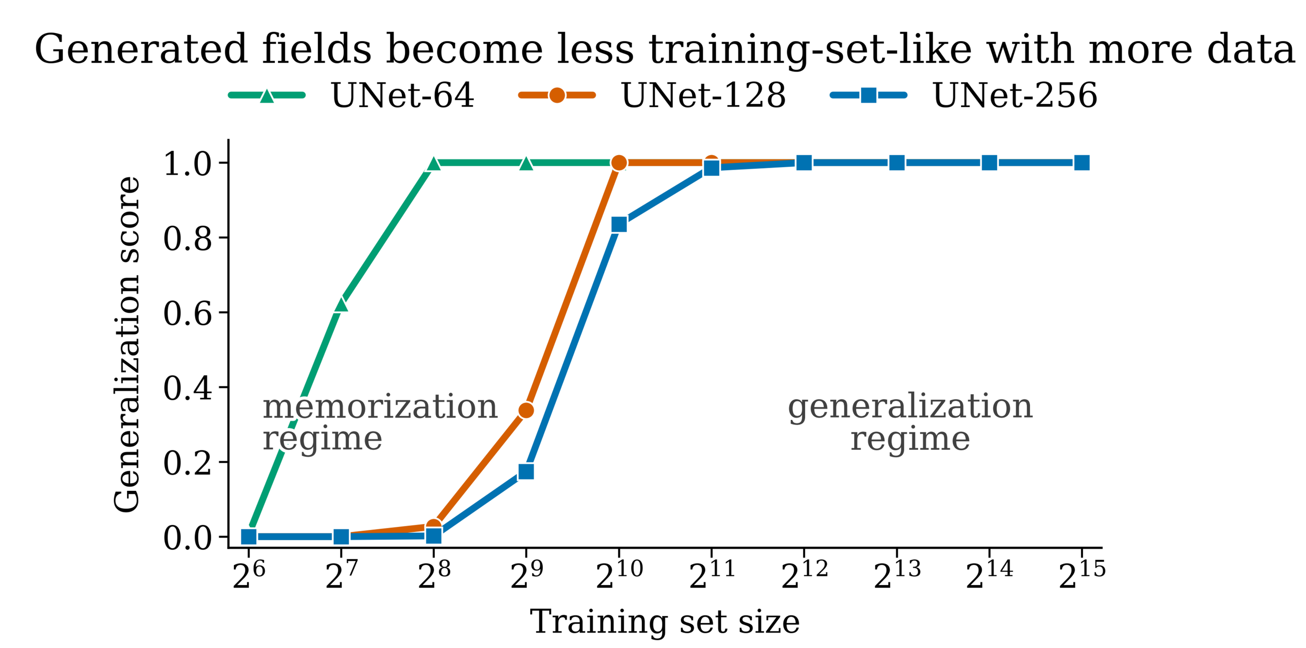


Calibration: does the generator obey θ?

Left to right, a cosmology vector $\boldsymbol{\theta}_{\rm ask}$ conditions the UNet through cross-attention while it denoises random noise into an HI map. The map is copied to 3 channels, resized, and encoded by frozen VGG16 features; an MLP head returns $\boldsymbol{\theta}_{\rm rec}$. The dashed loop asks whether $\boldsymbol{\theta}_{\rm rec} \approx \boldsymbol{\theta}_{\rm ask}$.



Memorization → generalization transition



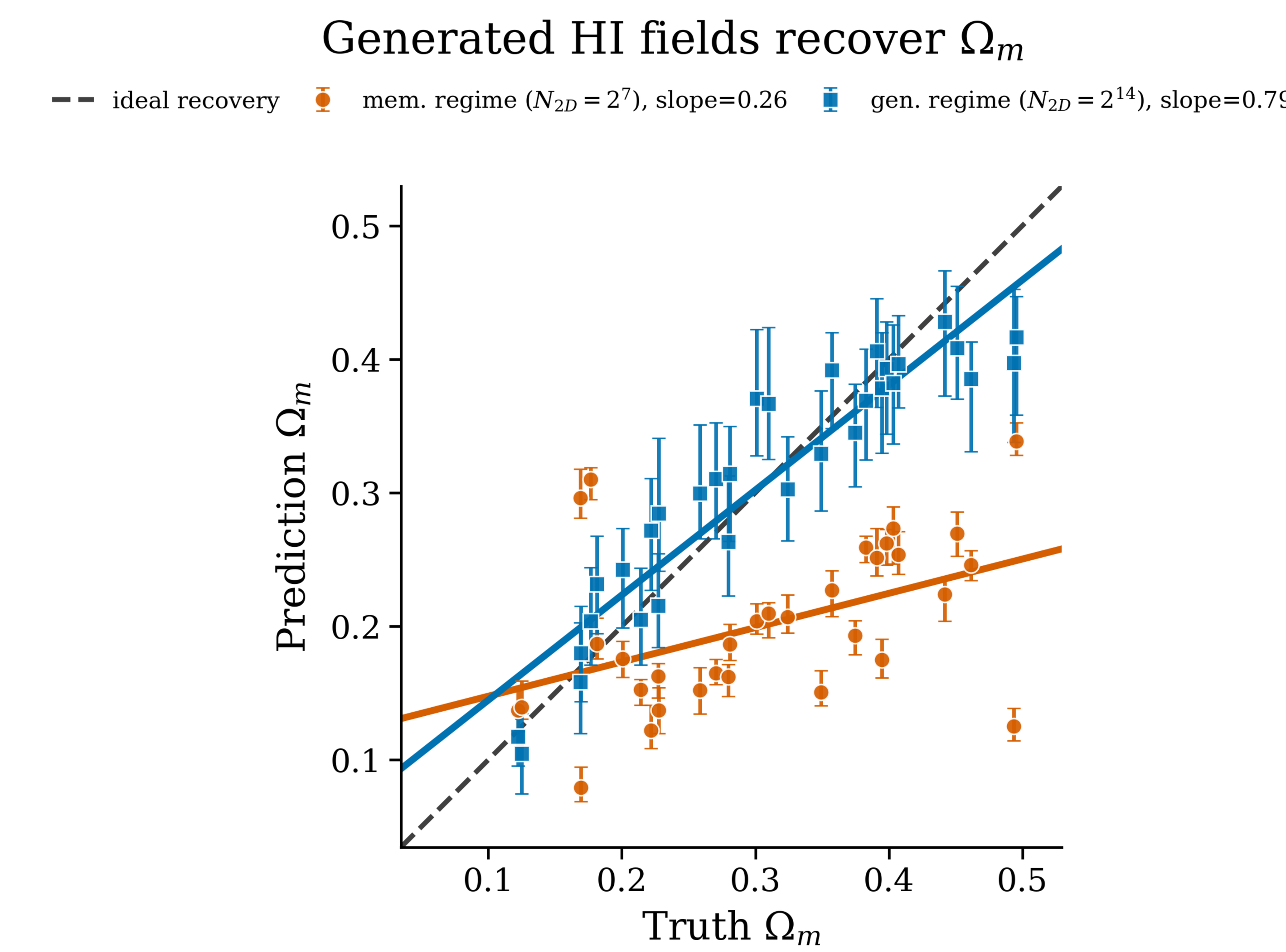
The generalization score is the fraction of generated maps that are *not* close to any training map,

$$\mathrm{GL} = \Pr[\, s_j < \tau \,],$$

with τ the 95th percentile of train-vs-train closest-match similarity (PCA space).

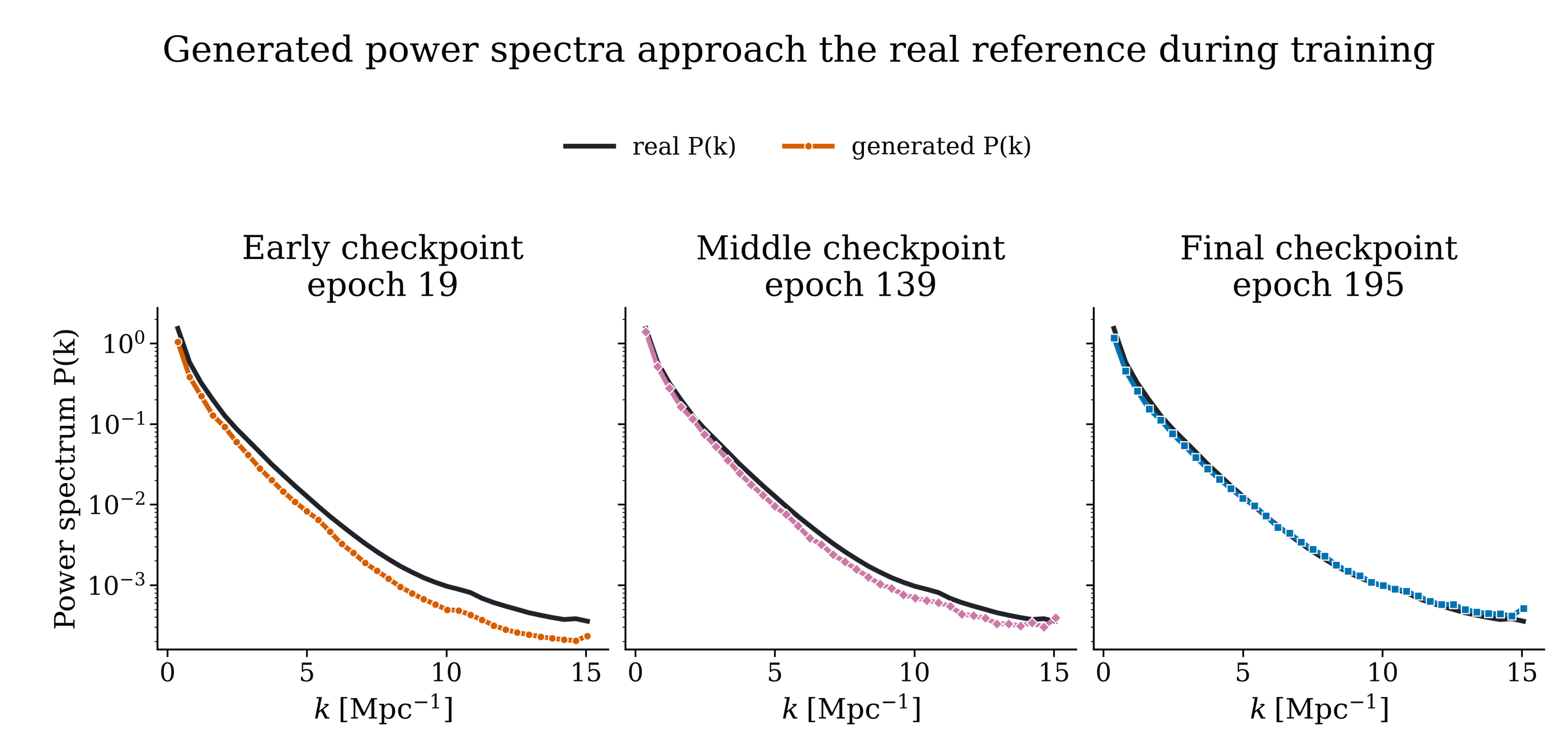
- $\mathrm{GL} \approx 0$: almost every generated map sits on top of a training map. The model is **copying**.
- $\mathrm{GL} \approx 1$: generated maps are no closer to training than training maps are to each other; the model makes **new fields**.
- The jump happens over a narrow range of N_{2D}; the wider UNet-256 needs more data (Zhang et al. 2024; Kadkhodaie et al. 2024; Yoon et al. 2023).

Calibration results



Main result: the bars show 16–84% seed scatter at fixed $\boldsymbol{\theta}_{\rm ask}$. In the high-data regime, the medians usually bracket the requested Ω_m within that scatter. In the memorization regime, seeds can return similar training-like maps, so the bars look tight while the median is biased. The slope (0.26 vs 0.79) supports the same point, but uncertainty calibration is the main message.

Physical fidelity



Across a single run, generated power spectra move toward the real reference from early to final checkpoints. Early checkpoints suppress power across much of $k\,[\mathrm{Mpc}^{-1}]$; by the final checkpoint the generated $P(k)$ follows the real shape much more closely.

Takeaways

1. **The transition appears in cosmological data.** The memorization-to-generalization transition known from natural images also occurs in CAMELS HI maps.
2. **It explains the error bars.** Small training sets give near-copies: inference looks precise but is biased toward the training values. Past the transition, $\boldsymbol{\theta}$ follows the input with honest scatter.
3. **It gives a practical check.** Nearest-neighbor similarity and encoder calibration reveal an emulator's regime before inference.